

Tracing the Hummingbird “One-to-Four” Rule Back to the Science

The most important correction

You were right to be skeptical of my earlier framing. **The interesting story is not a broken citation.** DOI 10.1525/cond.2008.8621 really does belong to McCaffrey & Wethington’s 2008 *Condor* paper, *How the Presence of Feeders Affects the Use of Local Floral Resources by Hummingbirds*. ¹

The genuinely interesting problem is **claim mutation**: a reasonable husbandry convention—roughly 20% sugar, or one part sugar to four parts water—gradually acquires a stronger scientific meaning than the underlying research warrants.

A concrete citation chain

One especially revealing chain looks like this:

Earlier field studies → McCaffrey & Wethington 2008 → Juntunen 2023 → public-facing “ideal ratio” language

McCaffrey & Wethington were **not studying optimal sugar concentration at all**. Their question was **whether hummingbirds changed their use of flowers when feeders were present**. For their experiment they simply maintained feeders at **1:4 sugar:water**, citing Kodric-Brown & Brown (1978) and Hilton & Miller (2003). ² Those earlier papers were **likewise using sugar water as an experimental tool**; McCaffrey cites them for the **method**, not for evidence that 1:4 is physiologically optimal. ³

Later in McCaffrey’s discussion, the authors call 1:4 a **“typical”** feeder solution, approximately 20%, and say that many hummingbird-pollinated flowers have nectar at equal or higher concentrations, citing Stiles & Freeman’s measurements of natural nectar and Roberts’s preference work. ⁴

Then comes the significant distortion. **Johanna Juntunen’s 2023 hummingbird-feeding report says that conservation organizations and scientists “agree that the ideal ratio” is 1:4 and cites, among other sources, McCaffrey & Wethington 2008.** ⁵ That is demonstrably stronger than what McCaffrey established. We have gone from:

“We used 1:4; it is typical and biologically plausible” → “this paper supports 1:4 as the scientifically agreed ideal.”

That is exactly the propagation phenomenon you were looking for.

The experimental literature actually says something messier

The research does **not** identify 20% as a universal hummingbird optimum. Blem et al. (2000), the paper I previously **overemphasized**, offered Rufous Hummingbirds solutions from 10–70%. When alternatives differed by 10 percentage points, **the birds preferred about 50%**, not 20%; around 20%, what increased was their ability to discriminate very small concentration differences. ⁶ So Blem is **not the origin of the 1:4 rule**, and “20% was their preferred concentration” would be a serious misreading.

Other foundational work likewise concerns tradeoffs rather than a magic number. Hainsworth & Wolf investigated nectar concentration and hummingbird energetics; Tamm & Gass studied energy-intake rates and concentration preferences; **Roberts showed that measured preference changes with feeding volume and time scale**. ⁷ Natural flowers themselves vary considerably: Stiles & Freeman measured nectar across 112 Costa Rican bird-visited species rather than discovering a single canonical concentration. ⁸

So the underlying “ground truth” is a **distributed body of physiology, behavioral ecology, and floral-nectar research**, not one experiment in which scientists determined that hummingbirds ought to receive exactly 20%.

How it becomes common knowledge

This helps explain what you encountered in Google. **Smithsonian's National Zoo currently gives the uncomplicated instruction 1 part sugar : 4 parts water**. ⁹ Cornell similarly recommends that recipe, while its more detailed guidance allows stronger or weaker mixtures under particular circumstances—evidence that 1:4 is a practical default rather than a biological constant. ¹⁰

The Sheri Williamson trail is even more revealing. *Birds & Blooms* names Williamson's *A Field Guide to Hummingbirds of North America* as a source. ¹¹ Williamson herself describes 1:4 as the **basic recipe**, but says solutions from roughly **1:3 through 1:6** can be appropriate depending on conditions. ¹² Older outreach material citing Williamson, Stokes, and other hummingbird authorities likewise presents 1:4 simply as accepted feeder practice. ¹³

What I think actually happened

The evidence supports **1:4 as a well-established, conservative husbandry rule of thumb**. It does *not* support the stronger proposition that “science discovered 20% is the optimal concentration for hummingbirds.” ¹⁴

And there really is a traceable semantic escalation:

experimental convention (“we used 1:4”) → ecological justification (“1:4 is typical and falls within natural nectar concentrations”) → expert practice (“use 1:4”) → scholarly overstatement (“scientists agree this is the ideal ratio”) → search/AI answer (“the correct ratio is 1:4”). ¹⁵

That is much more interesting than a miscitation. **Nobody necessarily fabricated the rule; each layer compressed away a qualification**. By the time Google presents it, a contingent, practical consensus has become something that looks like a single experimentally established fact.

1 How the Presence of Feeders Affects the Use of Local Floral Resources by Hummingbirds: A Case Study from Southern ArizonaEfectos de la Presencia de Alimentadores en el Uso de Recursos Florales Locales por Colibríes: Un Estudio de Caso en el Sur de ArizonaShort CommunicationsShort Communications | Ornithological Applications | Oxford Academic

https://academic.oup.com/condor/article/110/4/786/5152365?utm_source=chatgpt.com

2 3 4 15 (PDF) How the Presence of Feeders Affects the Use of Local Floral Resources by Hummingbirds: A Case Study from Southern Arizona

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/250068966_How_the_Presence_of_Feeders_Affects_the_Use_of_Local_Floral_Resources_by_Hummingbirds_A_Case_Study_from_Southern_Arizona)

[250068966_How_the_Presence_of_Feeders_Affects_the_Use_of_Local_Floral_Resources_by_Hummingbirds_A_Case_Study_from_Southern_Arizona](https://www.researchgate.net/publication/250068966_How_the_Presence_of_Feeders_Affects_the_Use_of_Local_Floral_Resources_by_Hummingbirds_A_Case_Study_from_Southern_Arizona)

5 Johanna Juntunen_Food Preference by Hummingbirds Visiting Feeders_2023

https://thegottliebnativegarden.com/wp-content/uploads/Johanna-Juntunen_Food-Preference-by-Hummingbirds-Visiting-Feeders_2023.pdf

6 oup.silverchair-cdn.com

<https://academic.oup.com/condor/article-abstract/102/1/235/5562890>

7 Crop volume, nectar concentration and hummingbird ...

https://www.sciencedirect.com/?utm_source=chatgpt.com

8 Patterns in Floral Nectar Characteristics of Some Bird-Visited Plant ...

https://www.jstor.org/stable/2389183?utm_source=chatgpt.com

9 14 Hummingbird Nectar Recipe

https://nationalzoo.si.edu/?utm_source=chatgpt.com

10 Here's What to Feed Your Summer Bird Feeder Visitors

https://www.allaboutbirds.org/?utm_source=chatgpt.com

11 Hummingbird Food: Sugar Water for Hummingbirds 101

https://www.birdsandblooms.com/birding/birding-basics/sugar-water-101/?utm_source=chatgpt.com

12 Uncategorized - Sheri L. Williamson

https://fieldguidetohummingbirds.com/category/uncategorized/?utm_source=chatgpt.com

13 Journey North Habitat Project: Unpave the Way for Wildlife

https://archive.journeynorth.org/unpave/hummer.html?utm_source=chatgpt.com